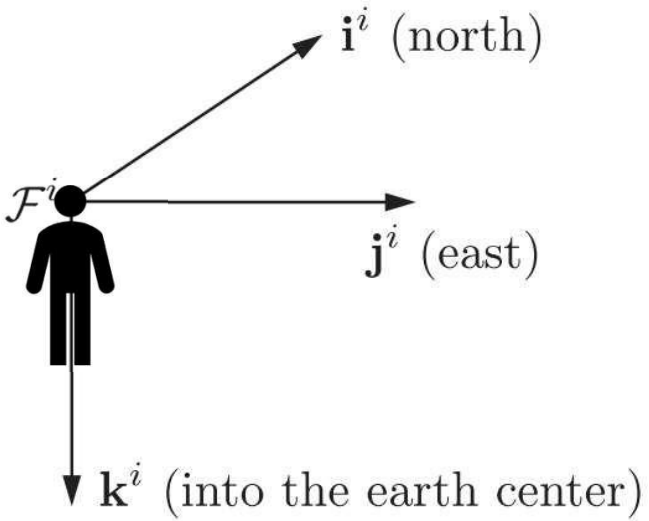
# EEEE4138 Aerial Robotics
# Tutorial 3 Forces and Moments




# Forces and Moments Tutorial




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## Question 1

A rotor runs at $\Omega = 500$ rad/s. Assume constant coefficients and constant rotor speed at this instant.

$$T = k_T \Omega^2 \qquad Q = k_M \Omega^2$$

$k_T = 2.00 \times 10^{-5}$ N/(rad/s)²
$k_M = 4.00 \times 10^{-7}$ N m/(rad/s)²

(a) Calculate thrust T and reaction-torque magnitude Q.
(b) Eliminate Ω to express Q in terms of T. State the ratio's units.
(c) If Ω doubles, how do T and Q change?
(d) Is this reaction torque the same as a gyroscopic moment?

## Solution 1: thrust and reaction torque

$T = (2.00 \times 10^{-5})(500)^2 = 5.00$ N

$Q = (4.00 \times 10^{-7})(500)^2 = 0.100$ N m

$\Omega^2 = T/k_T$, so $Q = (k_M/k_T)T = k_{MT}\,T$.

$k_{MT} = 0.0200$ m, because (N m)/N = m.

At twice the speed: T = 20.0 N and Q = 0.400 N m.

Reaction torque balances the drive against rotor drag. A gyroscopic moment arises when the body tilts the rotor angular momentum. They are different effects.

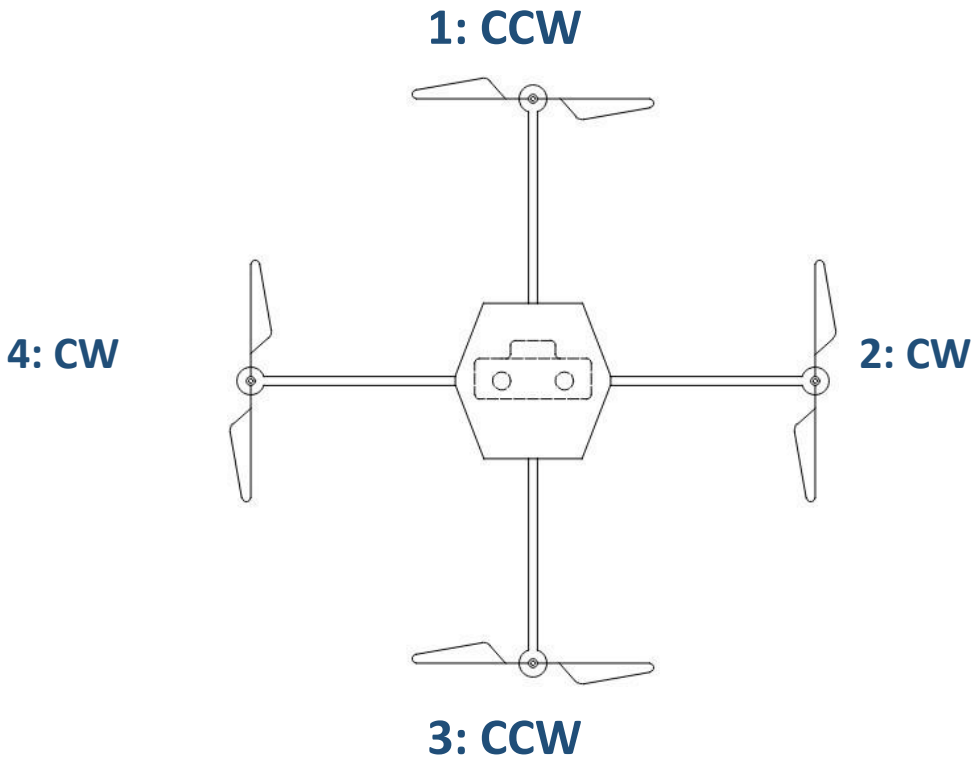


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## Question 2: + Quadrotor geometry

Rotor positions:
1 front: (l, 0, 0)
2 right: (0, l, 0)
3 rear: (−l, 0, 0)
4 left: (0, −l, 0)

l is centre-to-rotor distance.



## Question 2: Force and moment derivation

Use the geometry on the previous slide. Let $T_i \geq 0$ and let the airframe reaction-torque signs be $\sigma = [+1, -1, +1, -1]$.

$$r_i = [x_i, y_i, 0]^T \qquad F_i = [0, 0, -T_i]^T$$

$$M_i = r_i \times F_i + \sigma_i k_{MT} T_i e_z \qquad e_z = [0, 0, 1]^T$$

(a) Expand $r_i \times F_i$. Hence derive $F_z$, $M_x$, $M_y$ and $M_z$ for all four rotors.
(b) With $T_i = 6.00$ N for every rotor, calculate these four outputs.
(c) Increase $T_1$ by $\delta$ and decrease $T_3$ by $\delta$. Find the changes in total thrust, pitch moment and yaw moment.

## Solution 2: cross product and total loads

$$r_i \times F_i = [x_i, y_i, 0]^T \times [0, 0, -T_i]^T$$

$$= [-y_i T_i, x_i T_i, 0]^T$$

$$F_x = F_y = 0, \quad F_z = -(T_1 + T_2 + T_3 + T_4)$$

$$M_x = -lT_2 + lT_4 = l(T_4 - T_2)$$

$$M_y = lT_1 - lT_3 = l(T_1 - T_3)$$

$$M_z = k_{MT}(T_1 - T_2 + T_3 - T_4)$$



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## Solution 2: equal thrust and pitch change

With every $T_i = 6.00$ N:

$F_z = -24.0$ N, $M_x = M_y = M_z = 0$.

This is propulsive force only. Gravity is separate.

For $\Delta T_1 = \delta$, $\Delta T_3 = -\delta$, $\Delta T_2 = \Delta T_4 = 0$:

$\Delta F_z = -(\delta - \delta) = 0$

$\Delta M_y = l[\delta - (-\delta)] = 2l\delta$

$\Delta M_x = 0$, $\Delta M_z = k_{MT}(\delta - \delta) = 0$.



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## Question 3: Allocation and motor speeds

Use the + configuration and equations derived in Q2.
$l = 0.250$ m, $k_{MT} = 0.0200$ m,
$k_T = 2.00 \times 10^{-5}$ N/(rad/s)$^2$. Each rotor has $T_{max} = 8.00$ N.

The required propulsive outputs are:
$F_z = -24.0$ N, $M_x = 0$, $M_y = +0.500$ N m, $M_z = 0$.

(a) Solve for $T_1$, $T_2$, $T_3$ and $T_4$.
(b) Calculate each rotor speed using $\Omega_i = \sqrt{(T_i/k_T)}$.
(c) Check the rotor thrust limits.
(d) Keep $F_z$, $M_x$ and $M_z$ unchanged, but request $M_y = +2.00$ N m. Is this command feasible?

## Solution 3: solve for rotor thrusts

$M_x = 0$ gives $T_4 = T_2$.

$M_z = 0$ gives $T_1 + T_3 = T_2 + T_4$.

$-F_z = 24.0$ N gives each pair total = 12.0 N.

$M_y/l = 0.500/0.250 = T_1 - T_3 = 2.00$ N.

$T_1 = (12 + 2)/2 = 7.00$ N

$T_3 = (12 - 2)/2 = 5.00$ N

$T_2 = T_4 = 6.00$ N.



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## Solution 3: speeds and limits

$\Omega_i = \sqrt{(T_i/k_T)}$, using the nonnegative root.

$\Omega_1 = \sqrt{(7/0.0000200)} = 592$ rad/s

$\Omega_2 = \Omega_4 = \sqrt{(6/0.0000200)} = 548$ rad/s

$\Omega_3 = \sqrt{(5/0.0000200)} = 500$ rad/s

All thrusts lie between 0 and 8.00 N: feasible.

For $M_y = 2.00$ N m: $T_1 - T_3 = 8.00$ N.

$T_1 = 10.0$ N, $T_3 = 2.00$ N, $T_2 = T_4 = 6.00$ N.

$T_1$ exceeds 8.00 N: the requested outputs are infeasible.



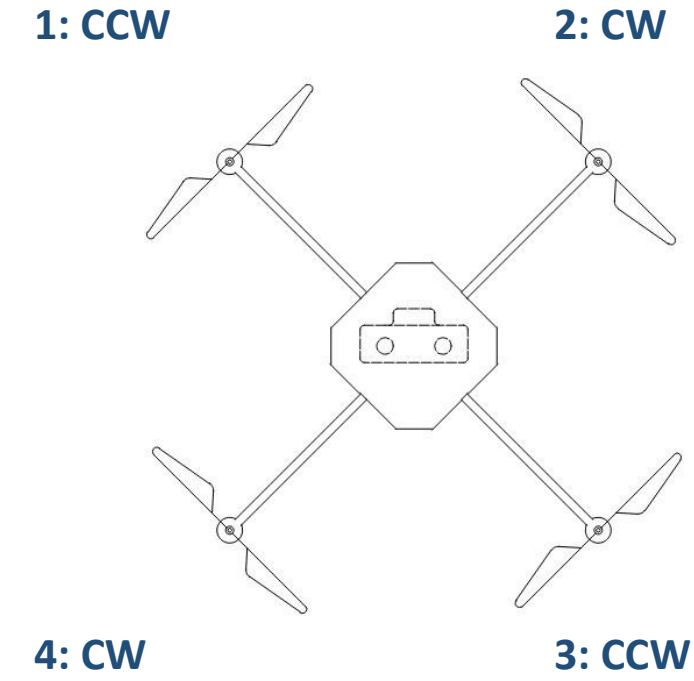
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## Question 4: X Quadrotor

Each arm has length l.
Let d = l/√2.

1 front-left: (d, −d, 0)
2 front-right: (d, d, 0)
3 rear-right: (−d, d, 0)
4 rear-left: (−d, −d, 0)

(a) Derive $M_x$, $M_y$ and $M_z$.
(b) Why is the lever arm d rather than l?
(c) Increase both front thrusts by δ and decrease both rear thrusts by δ. Which outputs change?




Top view. Front upwards.
z into page.

## Solution 4: X quadrotor

Each moment arm is $d = l/\sqrt{2}$, the perpendicular projection.

$M_x = d(T_1 - T_2 - T_3 + T_4)$

$M_y = d(T_1 + T_2 - T_3 - T_4)$

$M_z = k_{MT}(T_1 - T_2 + T_3 - T_4)$

For $\Delta T = [\delta, \delta, -\delta, -\delta]^T$:

$\Delta F_z = 0, \quad \Delta M_x = 0, \quad \Delta M_z = 0$

$\Delta M_y = d(\delta + \delta + \delta + \delta) = 4d\delta.$

Each rotor contributes roll and pitch, unlike the + layout.



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## Question 5: Fixed-wing lift and drag

A fixed-wing UAV flies steadily and level with thrust along the flight direction. Use m = 3.00 kg, ρ = 1.20 kg/m³ and wing area S = 0.500 m².

$L = \frac{1}{2}\rho V^2 S C_L$ $L = mg$

(a) Calculate the required $C_L$ at V = 10.0 m/s and 20.0 m/s.
(b) At 10.0 m/s, parasite drag is 2.00 N and induced drag is 8.00 N. Estimate each at 20.0 m/s and compare total drag.
(c) Explain why induced drag falls as speed rises in this case.

Assume unstalled flight and constant model coefficients.
At fixed lift: parasite drag $\propto V^2$, induced drag $\propto 1/V^2$.

## Solution 5: lift and drag

$L = mg = 3.00(9.81) = 29.43$ N.

$C_L = 2mg/(\rho V^2 S)$

At 10.0 m/s: $C_L = 29.43/30.0 = 0.981$.

At 20.0 m/s: $C_L = 29.43/120 = 0.245$.

Doubling speed multiplies parasite drag by 4:

$D_{parasite} = 8.00$ N. Induced drag divides by 4: 2.00 N.

Total drag is 10.0 N at both speeds in this example.

At fixed lift, higher speed needs a lower $C_L$, reducing induced drag.



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## Question 6: Hexarotor geometry (HOMEWORK)

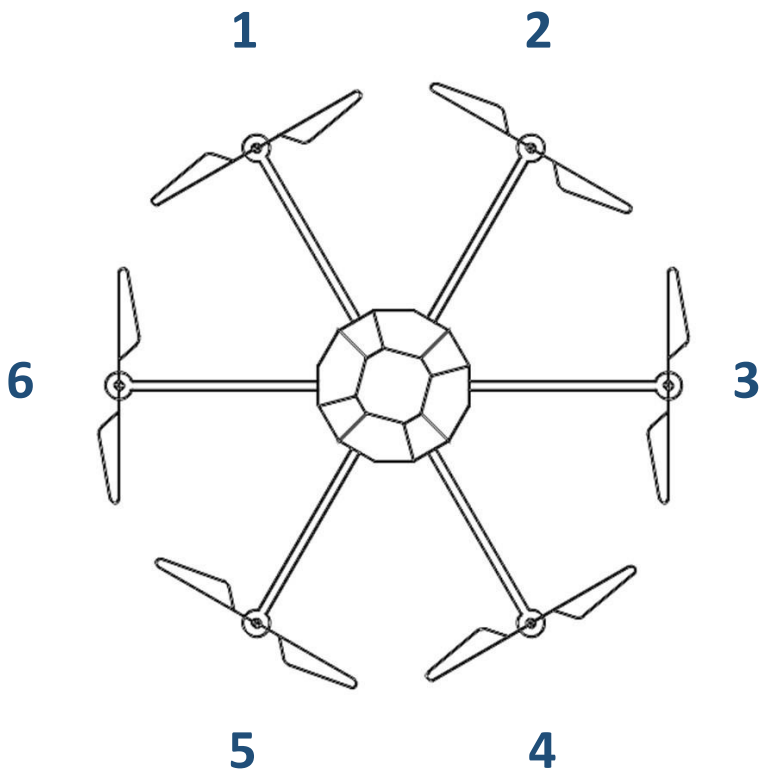
Six identical rotors on equal arms of length l.

Measure $\beta_i$ from +x towards +y:
$\beta = [-30°, 30°, 90°, 150°, 210°, 270°]$.

$x_i = l \cos \beta_i$
$y_i = l \sin \beta_i$

Odd rotors: CCW. Even rotors: CW.
$\sigma = [+1, -1, +1, -1, +1, -1]$.

Front upwards (between 1 and 2).
Top view: z into page.



## Question 6: Hexarotor allocation (HOMEWORK)

Use the positions and spin directions on the previous slide.

(a) Derive each rotor's contributions to $F_z$, $M_x$, $M_y$ and $M_z$.
(b) Assemble the 4 × 6 allocation matrix.
(c) Explain why an ordinary matrix inverse cannot give six rotor thrusts from four output commands.
(d) Find one allocation for total upward thrust 24.0 N and zero moments. Convert it to rotor speeds using $k_T = 2.00 \times 10^{-5}$ N/(rad/s)².

## Solution 6: six rotor contributions

Rotor i contributes $[-1, -y_i, x_i, \sigma_i k_{MT}]^T T_i$.

Let $a = l/2$, $b = \sqrt{3}\, l/2$ and $k = k_{MT}$.

$F_z = -(T_1 + T_2 + T_3 + T_4 + T_5 + T_6)$

$M_x = aT_1 - aT_2 - lT_3 - aT_4 + aT_5 + lT_6$

$M_y = b(T_1 + T_2 - T_4 - T_5)$

$M_z = k(T_1 - T_2 + T_3 - T_4 + T_5 - T_6)$

The coefficients of $T_1,\ldots,T_6$ form each row of $A_{hex}$.



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## Solution 6: allocation and nonuniqueness

$A_{hex}$ has 4 rows and 6 columns, so no ordinary inverse.

For this geometry with $l > 0$ and $k_{MT} > 0$, its rank is 4.

One solution: $T_1 = \ldots = T_6 = 4.00$ N.

$\Omega_1 = \ldots = \Omega_6 = \sqrt{(4/0.0000200)} = 447$ rad/s.

This gives $F_z = -24.0$ N and zero moments.

Another solution is $[4.2, 3.9, 3.9, 4.2, 3.9, 3.9]^T$ N.

It gives the same four outputs. The solution is not unique.

A practical allocator chooses among feasible solutions while respecting rotor limits.



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